Function Operations (composition)

Composition (the fifth operation)

The composition *f* with the function *g* is $(f \circ g)(x) = f(g(x))$.

$g(x)$ becomes the input (domain) of $f(x)$.

** $(f \circ g)(x) \neq (g \circ f)(x)$

Example 3: Let $f(x) = 2x^2$ & $g(x) = x - 1$. Find A) $(f \circ g)(x)$, B) $(g \circ f)(x)$, C) $(f \circ f)(x)$ and state the domain of each.

A) $(f \circ g)(x) =$

a) Find $(f \circ g)(2)$

Domain = _____

B) $(g \circ f)(x) =$

b) Find $(g \circ f)(0)$

Domain = _____

C) $(f \circ f)(x) =$

c) Find $(f \circ f)(-1)$

Domain = _____